

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I B Sandhya(192311292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sandhya

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one.

Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug?

Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

- i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent?
- Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

ANSWERS

1. Water Jug Problem

Problem

We have:

- One **4-gallon jug**
- One **3-gallon jug**
- No measuring marks
- Unlimited water supply from a pump

Goal: Get exactly 2 gallons of water in the 4-gallon jug.

Initial State

(0, 0)

Where:

- First value = amount of water in 4-gallon jug
- Second value = amount of water in 3-gallon jug

Solution Steps

Step	Action	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug	0	3
2	Pour 3-gallon jug into 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water from 3-gallon jug into 4-gallon jug until 4-gallon jug is full	4	2
5	Empty the 4-gallon jug onto the ground	0	2
6	Pour the remaining 2 gallons from 3-gallon jug into 4-gallon jug	2	0

Final State

(2, 0)

Therefore, **exactly 2 gallons of water are obtained in the 4-gallon jug.**

State-Space Representation

Initial State: (0, 0)

Goal State: (2, 0)

The possible operations are:

1. Fill a jug completely.
2. Empty a jug.
3. Pour water from one jug into another until the source is empty or the destination is full.

2.Mars Rover Agent

A Mars Rover is an intelligent agent that explores the surface of Mars, collects information, analyzes samples, and sends useful data to Earth.

i. What are the percepts for this agent?

The **percepts** are the information received by the rover through its sensors and instruments.

Examples:

- Images and videos of the Martian surface
- Temperature
- Atmospheric pressure
- Soil composition
- Rock and mineral composition
- Chemical properties of samples
- Radiation levels
- Distance from obstacles
- Position and location
- Wheel movement and battery status
- Weather and environmental conditions

ii. Characterize the operating environment

The Mars Rover's environment can be characterized as follows:

Property	Characteristic
Observability	Partially observable
Determinism	Partially stochastic
Episodic/Sequential	Sequential
Static/Dynamic	Dynamic
Discrete/Continuous	Continuous
Number of agents	Single-agent
Known/Unknown	Initially unknown/partially known

Explanation

- **Partially observable:** The rover cannot see the entire Mars environment at once.
- **Stochastic:** Unexpected events such as dust storms or mechanical failures may occur.
- **Sequential:** Current actions affect future decisions and states.
- **Dynamic:** The environment can change due to weather and other conditions.
- **Continuous:** The rover moves through a continuous physical environment.
- **Single-agent:** The rover primarily operates independently, although it communicates with Earth.
- **Unknown:** The rover explores areas that may not have been previously Studied.

iii. What are the actions the agent can take?

The Mars Rover can perform the following actions:

1. Move forward.
2. Move backward.

3. Turn left.
4. Turn right.
5. Stop.
6. Navigate around obstacles.
7. Take photographs.
8. Collect rock and soil samples.
9. Drill into rocks.
10. Analyze collected samples.
11. Store scientific data.
12. Transmit data to Earth.
13. Recharge or conserve energy when required.

iv. How can we evaluate the performance of the agent?

The rover's performance can be evaluated based on:

- Number of scientific samples successfully collected.
- Accuracy of sample analysis.
- Distance successfully explored.
- Quality and quantity of images collected.
- Amount of useful scientific data transmitted.
- Ability to avoid obstacles.
- Efficient use of battery power.
- Successful completion of assigned missions.
- Minimum damage to the rover.
- Safety and reliability of navigation.

Performance Measure

A good Mars Rover should **maximize scientific discoveries and useful data while minimizing energy consumption, travel time, and risk of damage.**

v. Suitable agent architecture

The most suitable architecture is a **Model-Based Utility-Based Learning Agent.**

Why?

The Mars Rover operates in a **partially observable and uncertain environment.** It needs to:

- Maintain an internal model of its surroundings.
- Make decisions based on available information.
- Select actions that provide maximum scientific value.
- Handle unexpected situations.
- Learn from previous exploration experiences.

A **model-based agent** helps maintain knowledge about the environment, while a **utility-based agent** helps choose the best action based on factors such as scientific value, energy,

safety, and time. A **learning component** allows the rover to improve its performance from experience.

3.8-Queens Problem

Problem Definition

The **8-Queens problem** is a classic AI search problem. The objective is to place **8 queens on an 8×8 chessboard** such that **no two queens attack each other**.

Since a queen can attack another queen:

- Horizontally
- Vertically
- Diagonally

Therefore, no two queens should be in the same row, column, or diagonal.

Problem Formulation

1. Initial State

An empty chessboard with **0 queens placed**.

Initial State = Empty Board

2. State

A state represents a **partial or complete placement of queens** on the chessboard.

For example:

[1, 5, 8, 6, 3, 7, 2, 4]

This means:

- Queen in Column 1 $\rightarrow$ Row 1
 - Queen in Column 2 $\rightarrow$ Row 5
 - Queen in Column 3 $\rightarrow$ Row 8
 - and so on.
-

3. Actions

Place a queen in an empty row of the next available column, provided that the queen does not attack any previously placed queen.

4. Transition Model

The transition model describes what happens when an action is performed.

Current state + Place a valid queen → New state

For example:

$[1, 5] \rightarrow [1, 5, 8]$

A queen is added to the next column in a safe position.

5. Goal Test

The goal is reached when:

- All **8 queens** are placed.
- No two queens attack each other.

Goal State

One possible solution is:

$[1, 5, 8, 6, 3, 7, 2, 4]$

6. Path Cost

The cost can be considered as:

- **1 unit for each queen placement**

For a complete solution:

Path Cost = 8

In practice, the main objective is to find a valid configuration rather than minimize path cost.

Search Space Exploration

The 8-Queens problem can be solved using:

1. Breadth-First Search (BFS)

Explores all possible configurations level by level.

2. Depth-First Search (DFS)

Places queens one by one and backtracks when a conflict occurs.

3. Backtracking Search

A specialized DFS approach that removes a queen when it leads to an invalid configuration.

4. Hill Climbing

Starts with a configuration and continuously moves toward a configuration with fewer conflicts.

5. Genetic Algorithm

Uses a population of possible solutions and evolves them using selection, crossover, and mutation.

6. Local Search

Can be used to minimize the number of attacking queen pairs.

Recommended Strategy

Backtracking Search is commonly used because it places queens incrementally and immediately eliminates invalid configurations.

4.OLA Cab Booking Problem

Problem

A customer wants to travel from one location to another using an OLA Cab booking application. The customer can select different cab types such as:

- Mini
- Micro
- Sedan
- Shared
- Prime

The customer wants to select a suitable cab and reach the destination.

Type of Problem-Solving Agent

A **Goal-Based Agent** is suitable for this problem.

Reason

The main goal of the customer is:

Reach the destination successfully.

The agent considers:

- Current location
- Destination
- Available cab types
- Cost
- Travel time
- Customer preferences

If the customer wants to select the **best cab based on cost, comfort, and travel time**, a **Utility-Based Agent** is more suitable.

Best Answer

Utility-Based Problem-Solving Agent

Because the agent can compare different cab options and select the one that provides the highest utility based on:

- Cost
- Comfort
- Travel time
- Availability
- Customer preference

Pseudocode

START

Input current_location

Input destination

Display available cab types

1. Micro

2. Mini
3. Sedan
4. Shared
5. Prime

Get user's preference

For each available cab:

- Calculate estimated fare
- Calculate estimated travel time
- Check cab availability
- Calculate utility score

Select the cab with the highest utility score
based on customer's preference

Display selected cab type
Display estimated fare
Display estimated travel time

Ask customer for confirmation

IF customer confirms THEN

- Book the selected cab
- Assign driver
- Track cab location

IF cab reaches pickup location THEN

- Start trip
- END IF

IF destination is reached THEN

- End trip
 - Calculate final fare
 - Display payment options
- END IF

ELSE

- Cancel booking

END IF

STOP

Simple Flow

Enter Source & Destination



Find Available Cabs



Select Cab Type



Compare Cost, Comfort & Time



Choose Best Cab



Confirm Booking



Assign Driver



Travel to Destination



Complete Trip

4. Uniform Cost Search (UCS) for Logistics Network

Problem

A logistics company wants to find the **least-cost path** from a starting warehouse **S** to the destination warehouse **G**.

Each route has a cost based on:

- Fuel
- Travel time
- Transportation expenses

The appropriate algorithm is **Uniform Cost Search (UCS)**.

Important Note

The **weighted graph itself is missing** from the question you sent. So the exact answer depends on the edge costs in the graph. Below is the general UCS procedure.

Uniform Cost Search

UCS always expands the node with the **lowest cumulative path cost** from the starting node.

It uses a **Priority Queue**.

Evaluation Function

$$f(n)=g(n)+h(n) \quad f(n) = g(n) \quad f(n)=g(n)$$

Where:

- $g(n)$ = total cost from starting node **S** to node n .

UCS Pseudocode

UNIFORM-COST-SEARCH(Graph, Start, Goal)

Create a priority queue FRONTIER

Insert Start with path cost 0

Create an empty EXPLORED set

WHILE FRONTIER is not empty:

 Remove the node with the lowest path cost

 IF node = Goal:

 Return the path and total cost

 IF node is not in EXPLORED:

 Add node to EXPLORED

 FOR each successor of node:

 Calculate new cost

$\text{new_cost} = \text{current_cost} + \text{edge_cost}$

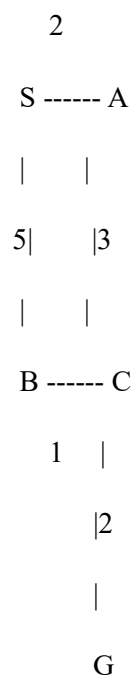
Add successor to FRONTIER

with new_cost

RETURN failure

Example

Suppose the network is:



Possible paths:

Path 1

$S \rightarrow A \rightarrow C \rightarrow G$

Cost:

$$2+3+2=7 \quad 2 + 3 + 2 = 7$$

Path 2

$S \rightarrow B \rightarrow C \rightarrow G$

Cost:

$$5+1+2=8 \quad 5 + 1 + 2 = 8$$

Therefore:

Optimal Path = $S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 7

UCS Expansion Example

	Step	Node Selected	Path	Cost
1	S	S		0
2	A	$S \rightarrow A$		2
3	C	$S \rightarrow A \rightarrow C$		5
4	G	$S \rightarrow A \rightarrow C \rightarrow G$	7	

Final Answer

Least-cost route:

$S \rightarrow A \rightarrow C \rightarrow G$

Total minimum cost: